

Laplace–Beltrami Formulation for Cell-Density Diffusion on a Crypt Surface

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Goal

- Start with a diffusion-type PDE posed in (x, y, z) on a crypt-shaped domain $\Omega \subset \mathbb{R}^3$.
- Re-parameterize the crypt surface by (θ, z) using a radius function $r(\theta, z)$ that encodes geometry.
- Write the transformed PDE using the **Laplace–Beltrami** operator
- State boundary conditions and a numerical solution outline.
- Coupe this to fission.

PDE in (x, y, z) (anisotropic, nonlinear diffusion)

Let $q = q(x, y, z, t)$ be (cell) density. Consider

$$\frac{\partial q}{\partial t} = \frac{\partial}{\partial x} \left(\frac{\alpha_x}{q^2} \frac{\partial q}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{\alpha_y}{q^2} \frac{\partial q}{\partial y} \right) + \frac{\partial}{\partial z} \left(\frac{\alpha_z}{q^2} \frac{\partial q}{\partial z} \right), \quad (x, y, z) \in \Omega.$$

Equivalently,

$$\frac{\partial q}{\partial t} = \nabla_x \cdot \left(\frac{1}{q^2} A \nabla_x q \right), \quad A = \text{diag}(\alpha_x, \alpha_y, \alpha_z).$$

Crypt surface parameterization

We define the crypt surface by

$$\mathbf{X}(\theta, z) = \begin{bmatrix} r(\theta, z) \cos(\theta) \\ r(\theta, z) \sin(\theta) \\ z \end{bmatrix}, \quad (\theta, z) \in [0, 2\pi] \times [0, L].$$

- $r(\theta, z)$ describes the crypt geometry (radius as a function of height and angle).
- In the **single crypt, axisymmetric** case: $r(\theta, z) = r(z)$ so $r_\theta = 0$.

Example: smooth “normal” crypt radius profile

A smooth profile (cylindrical midsection, tapered ends) is

$$r(z) = r_b (1 - e^{-az}) + r_t e^{a(z-L)},$$

where

- r_b = radius scale near the bottom,
- r_t = radius scale near the top opening,
- $a > 0$ controls how quickly the radii transition.

Compute the metric tensor g

Let $r = r(\theta, z)$. Compute tangent vectors:

$$\mathbf{X}_\theta = \begin{bmatrix} r_\theta \cos \theta - r \sin \theta \\ r_\theta \sin \theta + r \cos \theta \\ 0 \end{bmatrix}, \quad \mathbf{X}_z = \begin{bmatrix} r_z \cos \theta \\ r_z \sin \theta \\ 1 \end{bmatrix}.$$

Then the metric tensor g on the surface is

$$g = \begin{bmatrix} g_{\theta\theta} & g_{\theta z} \\ g_{z\theta} & g_{zz} \end{bmatrix} = \begin{bmatrix} \mathbf{X}_\theta \cdot \mathbf{X}_\theta & \mathbf{X}_\theta \cdot \mathbf{X}_z \\ \mathbf{X}_z \cdot \mathbf{X}_\theta & \mathbf{X}_z \cdot \mathbf{X}_z \end{bmatrix}.$$

Metric tensor entries, determinant, inverse

A direct computation gives

$$g_{\theta\theta} = r_\theta^2 + r^2, \quad g_{\theta z} = g_{z\theta} = r_\theta r_z, \quad g_{zz} = r_z^2 + 1.$$

Determinant:

$$|g| = (r_\theta^2 + r^2)(r_z^2 + 1) - (r_\theta r_z)^2 = r_\theta^2 + r^2(r_z^2 + 1).$$

Inverse metric:

$$g^{-1} = \frac{1}{|g|} \begin{bmatrix} r_z^2 + 1 & -r_\theta r_z \\ -r_\theta r_z & r_\theta^2 + r^2 \end{bmatrix}.$$

Nonlinear diffusion on the surface

To model diffusion along (θ, z) , define

$$\tilde{A} = \text{diag}(\alpha_\theta, \alpha_z).$$

The surface PDE for $q = q(\theta, z, t)$ in Laplace–Beltrami form is

$$\frac{\partial q}{\partial t} = \frac{1}{\sqrt{|g|}} \nabla_{\mathbf{u}} \cdot \left(\sqrt{|g|} \frac{1}{q^2} \tilde{A} g^{-1} \nabla_{\mathbf{u}} q \right), \quad \mathbf{u} = (\theta, z).$$

Boundary conditions (typical choices)

Periodicity in θ :

$$q(0, z, t) = q(2\pi, z, t), \quad \frac{\partial q}{\partial \theta}(0, z, t) = \frac{\partial q}{\partial \theta}(2\pi, z, t).$$

Bottom (no-flux) at $z = 0$:

$$\left(\sqrt{|g|} \frac{1}{q^2} w_z \right) \Big|_{z=0} = 0.$$

Top (prescribed density) at $z = L$:

$$q(\theta, L, t) = 1.$$

Numerical outline (pseudo-code)

Given initial condition $q(\theta, z, 0) = q_0(\theta, z)$ and geometry $r(\theta, z)$:

Time stepping

For $n = 0, 1, 2, \dots T$:

- 1 Compute r_θ, r_z and geometric terms $|g|, g^{-1}$.
- 2 Compute flux coefficients $\mathbf{w} = \tilde{A}g^{-1}\nabla_{\mathbf{u}}q$.
- 3 Advance

$$\frac{\partial q}{\partial t} = \frac{1}{\sqrt{|g|}} \nabla_{\mathbf{u}} \cdot \left(\sqrt{|g|} \frac{1}{q^2} \mathbf{w} \right) + H(z - z^*) \frac{\ln(2)}{T_c} q$$

- 4 Enforce boundary conditions (periodic in θ , no-flux at $z = 0$, Dirichlet at $z = L$).

Dynamic geometry via a fission factor $f(\theta, z; c)$

To incorporate fission, modify the surface parameterization:

$$\mathbf{X}(\theta, z; c) = \begin{bmatrix} f(\theta, z; c) r(\theta, z) \cos \theta \\ f(\theta, z; c) r(\theta, z) \sin \theta \\ z \end{bmatrix}.$$

A convenient choice is a smooth switch in z controlled by c :

$$\sigma(z; c) = \frac{1}{1 + e^{-\beta(z-c)}}, \quad f(\theta, z; c) = \sigma(z; c) + (1 - \sigma(z; c)) 2 \sin^2 \theta.$$

- For $z \ll c$: $f \approx 2 \sin^2 \theta$ (completely split profile).
- For $z \gg c$: $f \approx 1$ (unsplit profile).

Coupling the split parameter c to cell density

A simple coupling strategy:

- Define a scalar fission signal from q :

$$Q_f(t) = \frac{\int_{\Omega_p} q \, dS}{\int_{\Omega} q \, dS},$$

where Ω_p is a proliferative subregion of the surface (chosen by modeler).

- Evolve $c(t)$ with an ODE that keeps c in a bounded interval, e.g.

$$\frac{dc}{dt} = k Q_f(t) (c + c_0) (100 - c),$$